

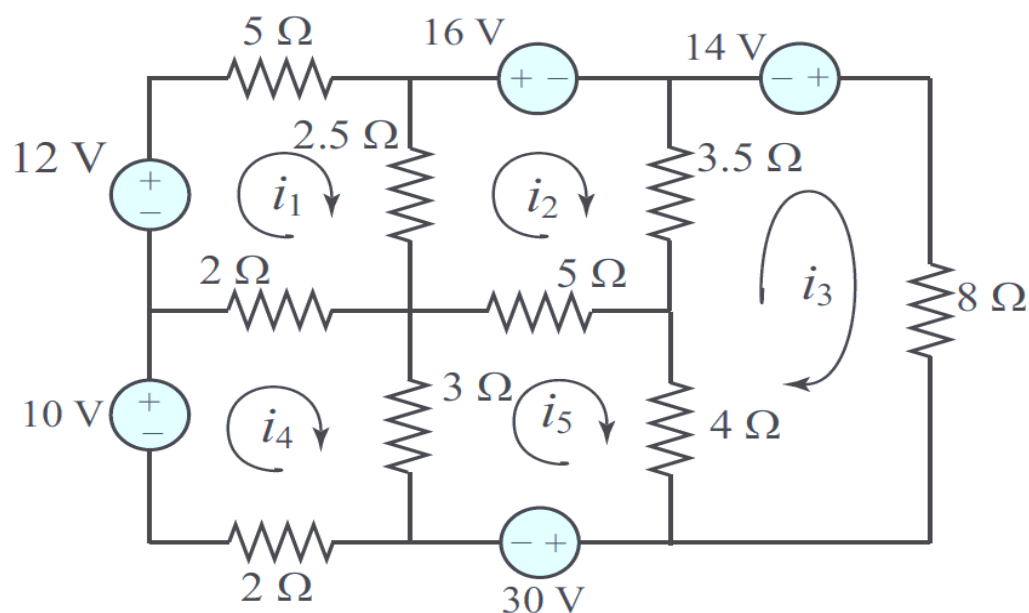
Solving a System of Linear Equations- Part 1

Example Problem

Analog Circuit Analysis (Mesh method)

Problem Statement

Analyze the current in each mesh in the given circuit and voltage sources.



Formulation of the Solution

The circuit is a linear system and we are asked to find 5 unknowns of $i_1 \sim i_5$.

There should be 5 *independent* equations to get the unique solution.

Applying Mesh current analysis for each meshes:

$$\begin{aligned}
 9.5i_1 - 2.5i_2 - 2i_4 &= 12 \\
 -2.5i_1 + 11i_2 - 3.5i_3 - 5i_5 &= -16 \\
 -3.5i_2 + 15.5i_3 - 4i_5 &= 14 \\
 -2i_1 + 7i_4 - 3i_5 &= 10 \\
 -5i_2 - 4i_3 - 3i_4 + 12i_5 &= -30
 \end{aligned}$$

To solve all the variables simultaneously, restructure the formula as **$\mathbf{Ax}=\mathbf{b}$** where $\mathbf{x}=[i_1 \ i_2 \ i_3 \ i_4 \ i_5]^T$. The matrix **$\mathbf{A}$** is a square matrix, and assume it is invertible.

$$\mathbf{Ax} = \mathbf{b}$$

$$\begin{bmatrix} 9.5 & -2.5 & 0 & -2 & 0 \\ -2.5 & 11 & -3.5 & 0 & -5 \\ 0 & -3.5 & 15.5 & 0 & -4 \\ -2 & 0 & 0 & 7 & -3 \\ 0 & -5 & -4 & -3 & 12 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} 12 \\ -16 \\ 14 \\ 10 \\ -30 \end{bmatrix}$$

Solve **$\mathbf{Ax}=\mathbf{b}$** for the variable vector **$\mathbf{x}$** to calculate $i_1 \sim i_5$

Solving with NM (MATLAB)

It is a system of linear equations (**$\mathbf{Ax}=\mathbf{b}$**) and the variable **$\mathbf{x}$** can be solved by using the matrix inverse of **$\mathbf{A}$** , if it is invertible.

Q. How can you check if the matrix A is invertible?

Use MATLAB **inv()** or **'\'** to solve for matrix inverse and vector **$\mathbf{x}$**

```
A=[ 9.5,-2.5,    0, -2,    0;
    -2.5, 11, -3.5,  0,   -5;
      0,-3.5, 15.5,  0,   -4;
     -2,   0,   0,  7,   -3;
      0,  -5,   -4, -3,  12];
```

```
b=[12; -16; 14; 10; -30];
```

```
xb=A\b
```

```
xbb=inv(A)*b
```

[Output] :

```
x = [0.1162; -3.9276; -1.1881; -0.5384; -4.6671]
```

Lecture Objective

We will learn how to solve a system of linear equations, $\mathbf{Ax}=\mathbf{b}$, by several approaches such as

- Gauss Elimination
- Gauss-Jordan Elimination
- LU decomposition
- Iterative Methods: Jacobi Iterative, Gauss-Seidel Iterative

Reading assignment

- Topics on Vectors, Matrix, Matrix Inverse, Gauss Elimination from any textbook of Linear Algebra.
- Recommendation
 - Textbook: Introduction to Linear Algebra by Gilbert Strang
 - OCW: [MIT OCW Linear Algebra 2011](#) by Gilbert Strang
 - YouTube: [Essence of linear algebra](#) by 3Blue1Brown

Introduction

Many engineering problems are involved in solving a system of linear equations, such as the linear circuit analysis as shown in the example problem.

A system of linear equations

A set of linear equations, which equations are in the form of

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b_n$$

- Each variable (x_i) is multiplied by a number, not another variable.
- Combining multiple linear equations, a linear system can be expressed as

$$\mathbf{Ax} = \mathbf{b}$$

$$\mathbf{A}(\mathbf{x} + \mathbf{y}) = \mathbf{Ax} + \mathbf{Ay}$$

$$\mathbf{A}(k\mathbf{x}) = k\mathbf{Ax}$$

- In engineering problems, each term can be interpreted as
 - $\mathbf{A}$: System characteristic matrix, $\mathbf{b}$: external or forced input, $\mathbf{x}$: variable
 - If $b = 0$ Homogeneous equation
 - If $b \neq 0$ Heterogeneous or Forced equation
 - Consider force equations we used in 'Statics' vs 'Dynamics'. ($F=ma$)

- A system of linear equations can be expressed as a linear combinations of vectors
 - Examples:

$$\mathbf{Ax} = c\mathbf{u} + d\mathbf{v} + e\mathbf{w} + f\mathbf{z} + \dots$$

- A linear system can use linear operators that satisfy superposition
 - Superposition:

$$af(x_1) + bf(x_2) = f(ax_1 + bx_2)$$

- Examples: differentiation, integration

$$f(\bullet) = \frac{d}{dt}(\bullet) , \quad f(\bullet) = \int (\bullet) dt$$

Q) Can you explain why a polynomial equation is not linear?

Before, we learn the numerical methods on solving linear systems, you need review the basic concept of vectors and matrices.

This lecture will cover

- Review on Vector/Matrix and Solving Linear Equations
 - Vector : dot/cross product, norm
 - Matrix: Multiplication, Transpose, Inverse, Types of Matrix(Square, Diagonal, Symmetric, Orthogonal, Triangular)
 - Rank
- Solving Linear Equations: $\mathbf{Ax} = \mathbf{b}$, $\mathbf{Ax} = \mathbf{0}$
 - Consistent / Inconsistent for homogeneous and heterogeneous system
- Gauss Elimination
 - Forward or back substitution with triangular matrix
 - Equivalent system : elementary operation
 - Pivoting
- Gauss-Jordan elimination
- LU Decomposition
- Solving for Matrix Inverse

Review Exercise

Write a brief definition for the following terminology:

- Vector
 - Dot product, cross product
 - Normal vector
 - Norm

- Matrix Rank:

The dimension of a matrix
= The maximum # of linearly independent rows/columns

- Square matrix
 - $A:(n \times n)$

- Symmetric matrix

A square matrix that is equal to its transpose

- Orthogonal / Orthonormal matrix

A square matrix whose columns & rows form orthonormal bases.
 $(A^T = A^{-1})$

- Hermitian – complex matrix
 - $A = A^* = \bar{A}^T$, Conjugate transpose.
- Unitary
 - $UU^* = U^*U = I$

- Positive definite*

- A symmetric matrix A where for any non-zero vector x , the result of the calculation $x^T A x$ is always a positive number.

- Similar matrix

Two square matrices A and B , that are related by an invertible matrix P s.t. $B = P^{-1} A P$.

Similar matrices represent the same transformation but under different bases and share key properties (eigen values, determinants...)

- Schur's Theorem*

- For any linear operator on a finite-dimensional complex vector space, there exists an orthonormal basis with respect to
- which the matrix of the operator is upper triangular.

Additional Notes:

- For $\mathbf{Ax}=\mathbf{b}$, where $A(m \times n)$, vector $\mathbf{b}(n \times 1)$, m : #eq, n : #var
 - If $m > n$, it is usually inconsistent,
 - But it could be consistent if $\text{rank}(A) == \text{rank}(A|\mathbf{b})$
 - If $m < n$, usually consistent with INF solution.
- Square matrix $\mathbf{A}(n \times n)$
 - Is it always invertible? No.
 - It can be singular (not invertible) if
 - $\text{rank}(A) < n$,
 - has 'zero' pivot
 - has eigenvalue of '0'
$$\det(A) = \lambda_1 \lambda_2 \dots \lambda_n, \quad (\text{if any eigenvalue is '0', the } \det(A)=0)$$
$$* \text{ trace}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n$$
- Every square matrix can be factored into upper triangular $\mathbf{T}$, as
$$\mathbf{A} = \mathbf{QTQ}^{-1}$$
 - If $\mathbf{A}$ is square and invertible, it reduces to row echelon form of $\mathbf{I}$
- Elimination does not preserve eigenvalue λ
 - but has the same determinant
- Real Symmetric matrix: $\mathbf{A} = \mathbf{A}^T$
 - always square
 - always diagonalizable
 - has only real eigenvalues
 - can have repeating eigenvalues, but always has orthogonal eigenvectors $\mathbf{Q}$. ($\mathbf{Q}^{-1} = \mathbf{Q}^T$)

Review Exercise

Solve the following examples by hand

P1. Solving with 2*2 Matrix: Solve $\mathbf{Ax}=\mathbf{b}$ with

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\begin{aligned} \mathbf{x} = \mathbf{A}^{-1}\mathbf{b} &= \frac{1}{\det \mathbf{A}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \\ &= \frac{1}{(a_{11}a_{22} - a_{12}a_{21})} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \end{aligned}$$

P2. Solving with 3*3 Matrix: Solve $\mathbf{Ax}=\mathbf{b}$ with

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 3 \\ -2 & 5 & -4 \\ 2 & -5 & 7 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 23 \\ 10 \\ -1 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} 5/12 & -11/18 & -19/36 \\ 1/6 & 2/9 & 1/18 \\ 0 & 1/3 & 1/3 \end{bmatrix} \begin{bmatrix} 23 \\ 10 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ 3 \end{bmatrix}$$

$\mathbf{Ax}=\mathbf{b}$

ANS $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \begin{bmatrix} 4 \\ 6 \\ 3 \end{bmatrix}$

$\mathbf{A}^{-1}$ by Gaussian Elimination

$$\begin{pmatrix} 2 & 1 & 3 & | & 1 & 0 & 0 \\ -2 & 5 & -4 & | & 0 & 1 & 0 \\ 2 & -5 & 7 & | & 0 & 0 & 1 \end{pmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$R_1+R_2, (-1) \times R_1+R_3$

$$\begin{pmatrix} 2 & 1 & 3 & | & 1 & 0 & 0 \\ 0 & 6 & -1 & | & 1 & 1 & 0 \\ 0 & -6 & 4 & | & -1 & 0 & 1 \end{pmatrix} \checkmark$$

R_2+R_3

$$\begin{pmatrix} 2 & 1 & 3 & | & 1 & 0 & 0 \\ 0 & 6 & -1 & | & 1 & 1 & 0 \\ 0 & 0 & 3 & | & 0 & 1 & 1 \end{pmatrix} \begin{matrix} \\ \\ -R_3+R_1 \end{matrix}$$

R_3+R_2

$$\begin{pmatrix} 2 & 1 & 0 & | & 1 & -1 & -1 \\ 0 & 6 & -1 & | & 1 & 1 & 0 \\ 0 & 0 & 1 & | & 0 & 1/3 & 1/3 \end{pmatrix}$$

P3. What is a general formula for solving $\mathbf{x}$ from $\mathbf{Ax}=\mathbf{b}$ with $\mathbf{A}:m \times m$, $\mathbf{b}:m \times 1$?

$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ · $\mathbf{A}$ is invertible

· The system has a unique solution.

$$\begin{pmatrix} 2 & 1 & 0 & | & 1 & -1 & -1 \\ 0 & 6 & -1 & | & 1 & 1 & 0 \\ 0 & 0 & 1 & | & 0 & 1/3 & 1/3 \end{pmatrix}$$

$-R_2+R_1$

$$\begin{pmatrix} 2 & 1 & 0 & | & 5/12 & -11/18 & -19/36 \\ 0 & 6 & -1 & | & 1 & 1 & 0 \\ 0 & 0 & 1 & | & 0 & 1/3 & 1/3 \end{pmatrix}$$

Solving Linear Equations (analytical)

For solving a system of linear equations, $\mathbf{Ax} = \mathbf{b}$, consider the following questions.

Q1. Does the system has a/the solution (consistent) or no solution (inconsistent)? *Consistent.*

Q2. If it is consistent, is the solution unique? *Yes*

Q3. Which method should we use to solve for ' $\mathbf{Ax}=\mathbf{b}$ '? *Gaussian Elimination*

Before we go on, you need to review:

What is the rank of a matrix? # of independent rows / columns

How to calculate the rank $\text{rank}(A) = \dim(\text{range}(A))$

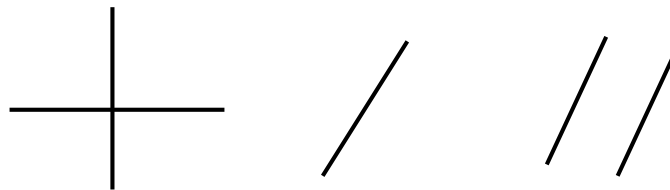
What is an independent and dependent equation?

If one equation can be expressed with linear combination of the variables of other equations, they are linearly dependent. If not, they are linearly independent

Q1+2. Does the system $\mathbf{Ax} = \mathbf{b}$ has a/the solution (consistent) or no solution(inconsistent)?
If it is consistent, then is the solution unique?(matrix $\mathbf{A}(m \times n)$, vector $\mathbf{b}(n \times 1)$)

**In this lecture, we will consider only square matrix A ($A: m * m$), unless stated otherwise. stated.*

For a linear system $\mathbf{Ax}=\mathbf{b}$, the solution can be either consistent or inconsistent as the below example.



For x_1, x_2 , a) unique sol. b) infinite sol c) no solution

Let's consider two cases, whether $\mathbf{b}=\mathbf{0}$ or $\mathbf{b} \neq \mathbf{0}$.

- $\mathbf{Ax}=\mathbf{0}$ (homogeneous, i.e., $\mathbf{b}=\mathbf{0}$) **is always consistent**

- Unique(Trivial) solution of $\mathbf{x}=\mathbf{0}$, if $\text{rank}(\mathbf{A}) = n$ _____
- Infinite solution including $\mathbf{x}=\mathbf{0}$, if $\text{rank}(\mathbf{A}) < n$ _____

rank(A|b)

- $Ax=b$, $b \neq 0$ (Heterogeneous)

- Inconsistent:

if $\text{rank}(A) < \text{rank}(A|b)$

Example:

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array} \right]. \text{ No solution.}$$

- Consistent:

if $\text{rank}(A) == \text{rank}(A|b)$

- Unique solution

if $\text{rank}(A) = \min(n, m)$

- Infinite solution

if $\text{rank}(A) < \min(n, m)$

Example:

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 0 \end{array} \right], \text{ Unique solutions}$$

$$[A|b] = \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{array} \right], \text{ Infinite solutions}$$

$(\text{rank}(A) = \text{rank}(A|b) = 2 < (n=3))$, free variables exist

- For unique solution: *Number of independent eq. == Number of variables*

By analyzing the matrix rank, we can determine if a heterogeneous system has the unique solution, infinite solution or no solution. *The matrix rank can be defined as the number of Independent row vectors.*

A method to find the matrix rank is converting the matrix A to 'row-echelon' matrix by row reduction using **Gauss-elimination**, which is the next lecture topic.

Q3. Which method should we use to solve for ' $Ax=b$ '?

The methods we will study are Gauss elimination, Gauss-Jordan elimination and LU decomposition

Gauss elimination

Background

Which expression is easier to find the solution? Which one is easier to count the number of independent equations?. The matrix **C** is an upper triangular matrix.

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \end{cases} \quad \text{vs} \quad \begin{cases} c_{11}y_1 + c_{12}y_2 + c_{13}y_3 = d_1 \\ c_{22}y_2 + c_{23}y_3 = d_2 \\ c_{33}y_3 = d_3 \end{cases}$$

Ax = b **Cy = d**

When a matrix is diagonal matrix, it is simpler to get the solution.

Example: **Cy=d**

$$\begin{aligned} c_{11}y_1 + 0 + 0 &= d_1 \\ c_{22}y_2 + 0 &= d_2 \\ c_{33}y_3 &= d_3 \end{aligned}$$

Q. Does a diagonal matrix always give a solution?

No

Example: Let $C(3,3)$ matrix in $Cy=d$ is a diagonal matrix with a '0' on its diagonal with $d_3 \neq 0, c_{33} = 0$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 3 \end{array} \right) \Rightarrow \text{rank}(C) < \text{rank}(C|d) \quad \text{No solution!}$$

Therefore, the purpose of Gaussian elimination is to transform the linear system into an equivalent form with an upper or lower triangular matrix:

$$\mathbf{A} \rightarrow \mathbf{U} \text{ or } \mathbf{A} = \mathbf{LU}$$

We will transform $Ax = b \rightarrow Cx = d$ such that **C** becomes a triangular matrix but it still gives the same solution **x**.

- $Ax = b$ is the equivalent form of $Cy = d$, if $x = y$ is satisfied.
- Making upper or lower triangular matrix is equivalent to making elements above or below the pivot element to be 'zero'
- Equivalent system using 'Elementary operations'
 - Multiply one equation by a constant
 - Interchange two equations
 - A linear combination with other equation: $c \times Eq(1) + d \times Eq(2) \leftarrow Eq(1)$

What is not preserved after elementary operation? How about the eigenvalues of matrix **A**?

Eigenvalues are not preserved.

Exercise:

Solve for $Ax=b$ using Gaussian elimination. (ANS: $x=[1,2,3]$)

$$A = \begin{bmatrix} 2 & 1 & 2 \\ 3 & 1 & 1 \\ 4 & 2 & 0 \end{bmatrix}, b = \begin{bmatrix} 10 \\ 8 \\ 8 \end{bmatrix}$$

$$x = A^{-1}b.$$

Gaussian elimination

$$\left(\begin{array}{ccc|ccc} 2 & 1 & 2 & 1 & 0 & 0 \\ 3 & 1 & 1 & 0 & 1 & 0 \\ 4 & 2 & 0 & 0 & 0 & 1 \end{array} \right) \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$$-1.5 \times R_1 + R_2 \quad (-3 \quad -1.5 \quad -3 \mid -1.5 \quad 0 \quad 0)$$

$$-2 \times R_1 + R_3 \quad (-4 \quad -2 \quad -4 \mid -2 \quad 0 \quad 0)$$

$$\left(\begin{array}{ccc|ccc} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & -0.5 & -2 & -1.5 & 1 & 0 \\ 0 & 0 & -4 & -2 & 0 & 1 \end{array} \right)$$

$$-R_2 + R_1 \quad (0 \quad -1 \quad -4 \mid -3 \quad 2 \quad 0)$$

$$\left(\begin{array}{ccc|ccc} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & 1 & 4 & 3 & -2 & 0 \\ 0 & 0 & -2 & -1 & 0 & 0.5 \end{array} \right)$$

$$2 \times R_3 + R_2 \quad (0 \quad 0 \quad -4 \mid -2 \quad 0 \quad 1)$$

$$-1 \times R_3 + R_1 \quad (0 \quad 0 \quad 2 \mid 1 \quad 0 \quad -0.5)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -0.5 & 1 & -0.25 \\ 0 & 1 & 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 0.5 & 0 & -0.25 \end{array} \right)$$

$$\therefore A^{-1} = \begin{bmatrix} -0.5 & 1 & -0.25 \\ 1 & -2 & 1 \\ 0.5 & 0 & -0.25 \end{bmatrix}$$

$$x = A^{-1}b = \begin{bmatrix} -0.5 & 1 & -0.25 \\ 1 & -2 & 1 \\ 0.5 & 0 & -0.25 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\therefore x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Solve for $\mathbf{Ax}=\mathbf{b}$ using Gaussian elimination. (ANS: $\mathbf{x}=[10,-7,-42]$)

$$\mathbf{C} = [\mathbf{A} \mid \mathbf{b}] = \left[\begin{array}{ccc|c} 2 & 1 & 2 & 10 \\ 3 & 1 & 1 & 8 \\ 4 & 4 & 0 & 6 \end{array} \right] = \left[\begin{array}{c} R_1 \\ R_2 \\ R_3 \end{array} \right]$$

$$\mathbf{C}^{(2)} = \begin{bmatrix} [2 \ 1 \ 2 \ 10] \\ [0 \ -0.5 \ -2 \ -7] \\ [0 \ 2 \ -4 \ -14] - \left(\frac{2}{-0.5}\right)[0 \ -0.5 \ -2 \ -7] = [0 \ 0 \ -12 \ -42] \end{bmatrix} = \begin{bmatrix} R_1^{(1)} \\ R_2^{(1)} \\ R_3^{(1)} - \left(\frac{2}{-0.5}\right)R_2^{(1)} \end{bmatrix}$$

$$= \left[\begin{array}{ccc|c} 2 & 1 & 2 & 10 \\ 0 & -0.5 & -2 & -7 \\ 0 & 0 & -12 & -42 \end{array} \right]$$

Example:

Revisit to the example problem in the beginning of the lecture (Mesh analysis to calculate current). Apply Gauss-elimination method to find the solution.

Let R_i is the equation at i^{th} row. Note: $\{ * \}$ is NOT a matrix, it is a set of equations

$$\left\{ \begin{array}{cccccc} 4x_1 & -2x_2 & -3x_3 & +6x_4 & = & 12 \\ -6x_1 & +7x_2 & +6.5x_3 & -6x_4 & = & -6.5 \\ x_1 & +7.5x_2 & +6.25x_3 & +5.5x_4 & = & 16 \\ -12x_1 & +22x_2 & +15.5x_3 & -x_4 & = & 17 \end{array} \right\} = \left\{ \begin{array}{c} R_1 \\ R_2 \\ R_3 \\ R_4 \end{array} \right\}$$

- Step 1) First pivot: Eliminate elements below the first pivot.

$$\left\{ \begin{array}{c} R_1' \\ R_2' \\ R_3' \\ R_4' \end{array} \right\} = \left\{ \begin{array}{c} R_1 \\ R_2 - \left(\frac{-6}{4} \right) R_1 \\ R_3 - \left(\frac{1}{4} \right) R_1 \\ R_4 - \left(\frac{-12}{4} \right) R_1 \end{array} \right\} = \left\{ \begin{array}{c} R_1 \rightarrow R_1 \\ R_2 \rightarrow R_2 - \frac{a_{21}}{a_{11}} R_1 \\ R_3 \rightarrow R_3 - \frac{a_{31}}{a_{11}} R_1 \\ R_4 \rightarrow R_4 - \frac{a_{41}}{a_{11}} R_1 \end{array} \right\}$$

$$\left\{ \begin{array}{cccccc} 4x_1 & -2x_2 & -3x_3 & +6x_4 & = & 12 \\ -6x_1 - \left(\frac{-6}{4} \right) 4x_1 & +7x_2 - \left(\frac{-6}{4} \right) (-2x_2) & +6.5x_3 - \left(\frac{-6}{4} \right) (-3x_3) & -6x_4 - \left(\frac{-6}{4} \right) 6x_4 & = & 11.5 \\ x_1 - \left(\frac{1}{4} \right) 4x_1 & +7.5x_2 - \left(\frac{1}{4} \right) (-2x_2) & +6.25x_3 - \left(\frac{1}{4} \right) (-3x_3) & +5.5x_4 - \left(\frac{1}{4} \right) 6x_4 & = & 13 \\ -12x_1 - \left(\frac{-12}{4} \right) 4x_1 & +22x_2 - \left(\frac{-12}{4} \right) (-2x_2) & +15.5x_3 - \left(\frac{-12}{4} \right) (-3x_3) & -x_4 - \left(\frac{-12}{4} \right) 6x_4 & = & 53 \end{array} \right\}$$

$$= \left\{ \begin{array}{cccccc} 4x_1 & -2x_2 & -3x_3 & +6x_4 & = & 12 \\ & 4x_2 & +2x_3 & 3x_4 & = & 11.5 \\ & 8x_2 & +7x_3 & +4x_4 & = & 13 \\ & 16x_2 & +6.5x_3 & +17x_4 & = & 53 \end{array} \right\}$$

- Step 2) Second pivot: Eliminate elements below the second pivot.

$$\begin{aligned}
 \begin{Bmatrix} R_1'' \\ R_2'' \\ R_3'' \\ R_4'' \end{Bmatrix} &= \begin{Bmatrix} R_1' \\ R_2' \\ R_3' - \left(\frac{8}{4}\right)R_2' \\ R_4' - \left(\frac{16}{4}\right)R_2' \end{Bmatrix} = \begin{Bmatrix} R_1' \rightarrow R_1' \\ R_2' \rightarrow R_2' \\ R_3' \rightarrow R_3' - \frac{a_{32}'}{a_{22}'}R_2' \\ R_4' \rightarrow R_4' - \frac{a_{42}'}{a_{22}'}R_2' \end{Bmatrix} \\
 &= \begin{Bmatrix} 4x_1 & -2x_2 & -3x_3 & +6x_4 & =12 \\ & 4x_2 & +2x_3 & 3x_4 & =11.5 \\ & & 3x_3 & -2x_4 & =-10 \\ & & -1.5x_3 & +5x_4 & =7 \end{Bmatrix}
 \end{aligned}$$

- Step 3) Third pivot

$$\begin{Bmatrix} R_1^{(3)} \\ R_2^{(3)} \\ R_3^{(3)} \\ R_4^{(3)} \end{Bmatrix} = \begin{Bmatrix} R_1'' \\ R_2'' \\ R_3'' \\ R_4'' - \left(\frac{6.5}{7}\right)R_3'' \end{Bmatrix} = \begin{Bmatrix} R_1'' \rightarrow R_1'' \\ R_2'' \rightarrow R_2'' \\ R_3'' \rightarrow R_3'' \\ R_4'' \rightarrow R_4'' - \left(\frac{a_{43}''}{a_{33}''}\right)R_3'' \end{Bmatrix}$$

$$= \begin{Bmatrix} 4x_1 & -2x_2 & -3x_3 & +6x_4 & =12 \\ & 4x_2 & +2x_3 & 3x_4 & =11.5 \\ & & 3x_3 & -2x_4 & =-10 \\ & & & 4x_4 & =2 \end{Bmatrix}$$

- Now, the system is in the equivalent form of a Triangular matrix.
- First, we can get the solution of $x_4 = 0.5$. Then, use the back-substitution to get the rest solution.

Back-substitution

$$\begin{aligned}
 a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 &= b_1 & a_{11}x_1 &= b_1 - (a_{12}x_2 + a_{13}x_3 + a_{14}x_4) \\
 a_{22}x_2 + a_{23}x_3 + a_{24}x_4 &= b_2 & \rightarrow a_{22}x_2 &= b_2 - (a_{23}x_3 + a_{24}x_4) \\
 a_{33}x_3 + a_{34}x_4 &= b_3 & a_{33}x_3 &= b_3 - (a_{34}x_4) \\
 a_{44}x_4 &= b_4 & a_{44}x_4 &= b_4 \\
 \therefore x_i &= \frac{1}{a_{ii}} \left(b_i - \sum_{j=i+1}^n a_{ij}x_j \right)
 \end{aligned}$$

- The solution in terms of coefficients becomes

$$x_4 = \frac{b_4}{a_{44}},$$

$$x_3 = \frac{1}{a_{33}} (b_3 - a_{34}x_4)$$

$$x_2 = \frac{1}{a_{22}} (b_2 - (a_{23}x_3 + a_{24}x_4))$$

$$x_1 = \frac{1}{a_{11}} (b_1 - (a_{12}x_2 + a_{13}x_3 + a_{14}x_4))$$

There are cases when unique solution is not obtainable with the elimination. You must check for the following cases in your algorithm for solving a linear system using Gaussian elimination

- Case 1. When there is a zero in the pivot in the row-reduced form
 - No solution. $\text{rank}(A) < \text{rank}(A|b)$
- Case 2. For n equations, number of pivots is less than n
 - Infinite solution. $\text{rank}(A) = \text{rank}(A|b) < n$
- Case 3. There is zero in the pivot position during the elimination process.
 - Use pivoting to exchange row equations → Next lecture topic

Algorithm

- $(A:m \times n)$, $b: m \times 1$
- m : number of rows (equations)
- n : number of columns, excluding b or $n+1$: including b

$$\begin{array}{ccccccc} & & & & & & 0 \\ & & & & & & 0 \\ & & & & & & 0 \\ & & & & & & 0 \\ & & & & & & 0 \\ & & & & & & 0 \\ & & & & & & 0 \\ & & & & & & 0 \\ & & & & & & 0 \\ & & & & & & 0 \end{array}$$

$k=1$

$$\left\{ \begin{array}{l} R_1 \rightarrow R_1 \\ R_2 \rightarrow R_2 - \frac{a_{21}}{a_{11}} R_1 \\ R_3 \rightarrow R_3 - \frac{a_{31}}{a_{11}} R_1 \\ R_4 \rightarrow R_4 - \frac{a_{41}}{a_{11}} R_1 \end{array} \right\}$$

$$\begin{aligned} a_{11} &= a_{11}, a_{12} = a_{12}, \dots, a_{1j} = a_{1j}, \dots \\ a_{21} &= a_{21} - \left(\frac{a_{21}}{a_{11}} \right) a_{11}, a_{22} = a_{22} - \left(\frac{a_{21}}{a_{11}} \right) a_{12}, \dots, a_{2j} = a_{2j} - \left(\frac{a_{21}}{a_{11}} \right) a_{1j}, \dots \\ a_{31} &= a_{31} - \left(\frac{a_{31}}{a_{11}} \right) a_{11}, a_{32} = a_{32} - \left(\frac{a_{31}}{a_{11}} \right) a_{12}, \dots, a_{3j} = a_{3j} - \left(\frac{a_{31}}{a_{11}} \right) a_{1j}, \dots \\ a_{ij} &\rightarrow a_{ij} - \left(\frac{a_{i1}}{a_{11}} \right) a_{1j} \end{aligned} \quad \begin{array}{l} k=1 \\ i=2 \text{ to } m \\ j=1 \text{ to } n \end{array}$$

$k=2$

$$\left\{ \begin{array}{l} R'_1 \rightarrow R'_1 \\ R'_2 \rightarrow R'_2 \\ R'_3 \rightarrow R'_3 - \frac{a'_{32}}{a'_{22}} R'_2 \\ R'_4 \rightarrow R'_4 - \frac{a'_{42}}{a'_{22}} R'_2 \end{array} \right\}$$

$$\begin{aligned} a_{11} &= a_{11}, a_{12} = a_{12}, \dots, a_{1j} = a_{1j}, \dots \\ a_{21} &= a_{21}, a_{22} = a_{22}, \dots, a_{2j} = a_{2j}, \dots \\ 0, a_{32} &= a_{32} - \left(\frac{a_{32}}{a_{22}} \right) a_{22}, a_{33} = a_{33} - \left(\frac{a_{32}}{a_{22}} \right) a_{23}, \dots, a_{3j} = a_{3j} - \left(\frac{a_{32}}{a_{22}} \right) a_{2j}, \dots \\ a_{ij} &\rightarrow a_{ij} - \left(\frac{a_{i2}}{a_{22}} \right) a_{2j} \end{aligned} \quad \begin{array}{l} k=2 \\ i=3 \text{ to } m \\ j=2 \text{ to } n \end{array}$$

$k=3$

$$\left\{ \begin{array}{l} R''_1 \rightarrow R''_1 \\ R''_2 \rightarrow R''_2 \\ R''_3 \rightarrow R''_3 \\ R''_4 \rightarrow R''_4 - \frac{a''_{43}}{a''_{33}} R''_3 \end{array} \right\}$$

$$\begin{aligned} a_{11} &= a_{11}, \dots, a_{1j} = a_{1j}, \dots \\ a_{21} &= a_{21}, \dots, a_{2j} = a_{2j}, \dots \\ a_{31} &= a_{31}, \dots, a_{3j} = a_{3j}, \dots \\ 0, 0, &= a_{43} - \left(\frac{a_{43}}{a_{33}} \right) a_{33}, a_{44} = a_{44} - \left(\frac{a_{43}}{a_{33}} \right) a_{34}, \dots \\ &, a_{4j} = a_{4j} - \left(\frac{a_{43}}{a_{33}} \right) a_{3j} \end{aligned}$$

$k=k$

repeat
until $k=m-1$

$$\begin{aligned} a_{ij} &\rightarrow a_{ij} - \left(\frac{a_{i3}}{a_{33}} \right) a_{3j} \\ k &= 3 \\ i &= 4 \text{ to } m \\ j &= 3 \text{ to } n \end{aligned}$$

Exercise:

Write a pseudocode

- You can either treat **A**, **b** separately or augment them together. ($m \times n$ or $m \times (n+1)$)
- It is preferred to separate **A**, **b**, for we may need to transform just the matrix **A**, such as LU decomposition.

HINT:
$$a_{ij} \rightarrow a_{ij} - \left(\frac{a_{ik}}{a_{kk}} \right) a_{kj}, \quad x_i = \frac{1}{a_{ii}} \left(b_i - \sum_{j=i+1}^n a_{ij} x_j \right)$$

(1). Row-echelon form using elimination

```
for k = 1 to (m-1)
  for i = k to m
    N = aik / akk
    for j = k to n
      aij = aij - N * akj
    end for
    bi = bi - N * bk
  end for
end for
```

(2) Back-substitution

```
for i = m to 1
  s = 0
  for j = (i+1) to n
    s = s + aij * xj
  end for
  xi = 1 / aii * (bi - s)
```

end for

Write a C-program

See assignment instructions

```
void gaussElim(matrix A, matrix b, matrix U, matrix d)

for (int k = 1; k < m; k++) {
    for (int i = k; i < (m+1); i++) {
        N = A[i][k] / A[k][k];
        for (int j = k; j < (n+1); j++) {
            U[i][j] = A[i][j] - N * A[k][j];
        }
        d[i] = b[i] - N * b[k];
    }
}
```

